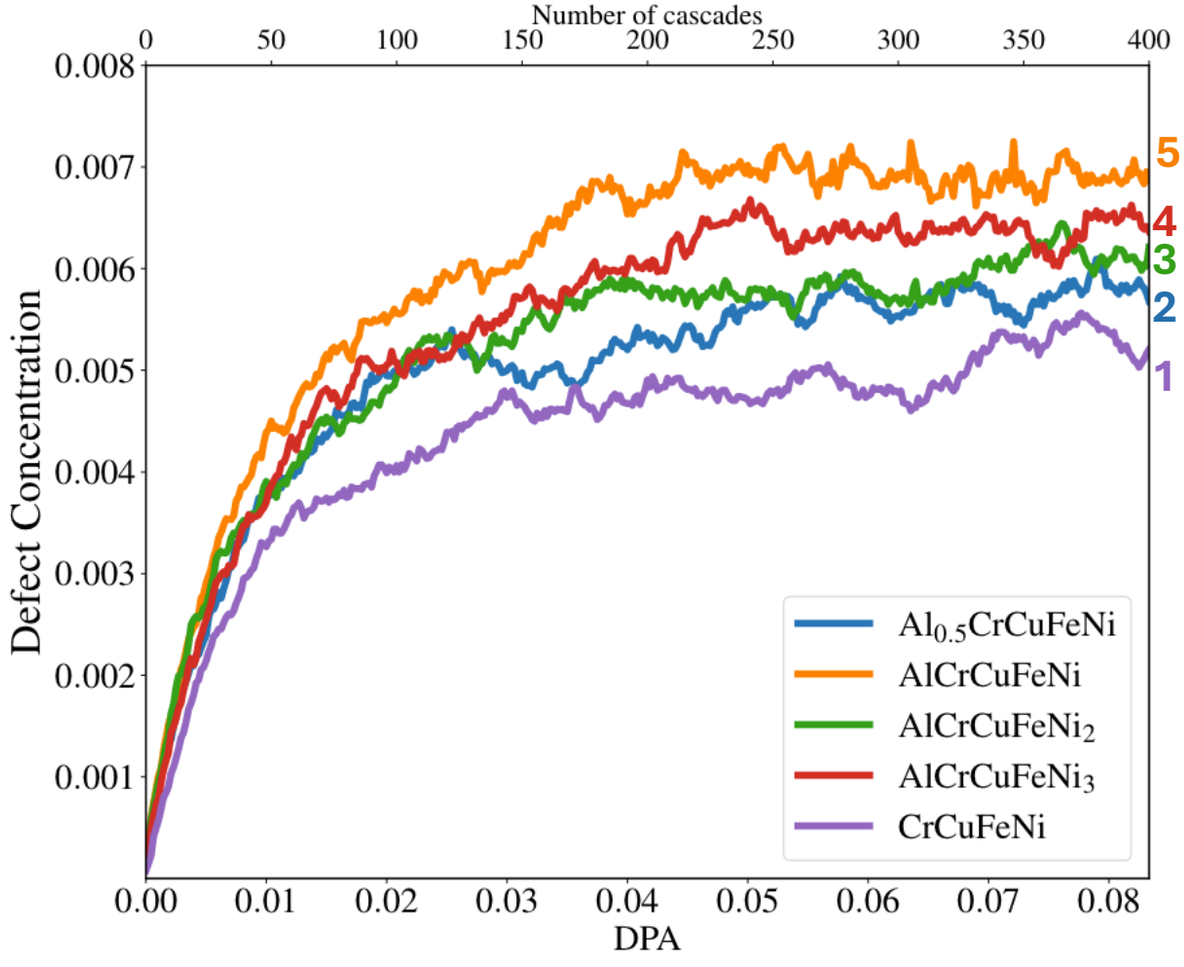
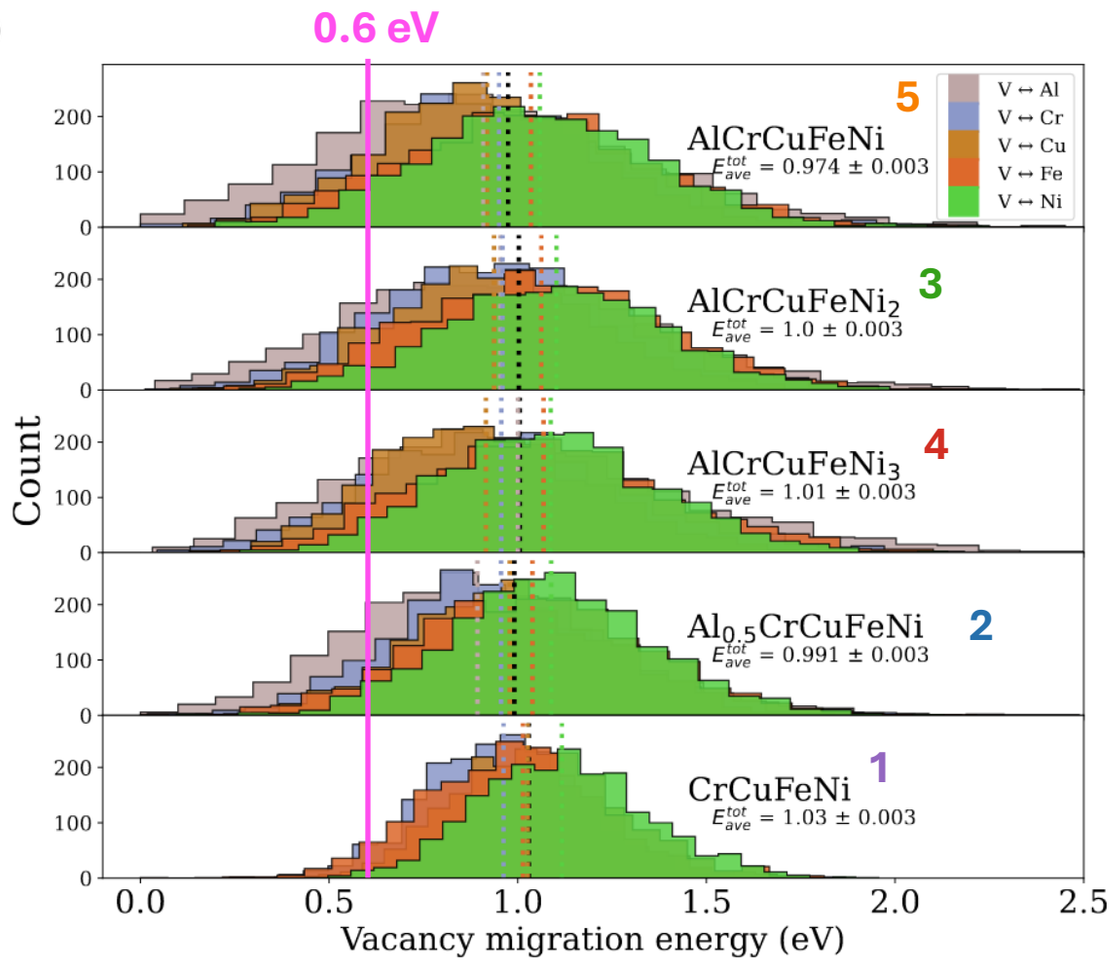
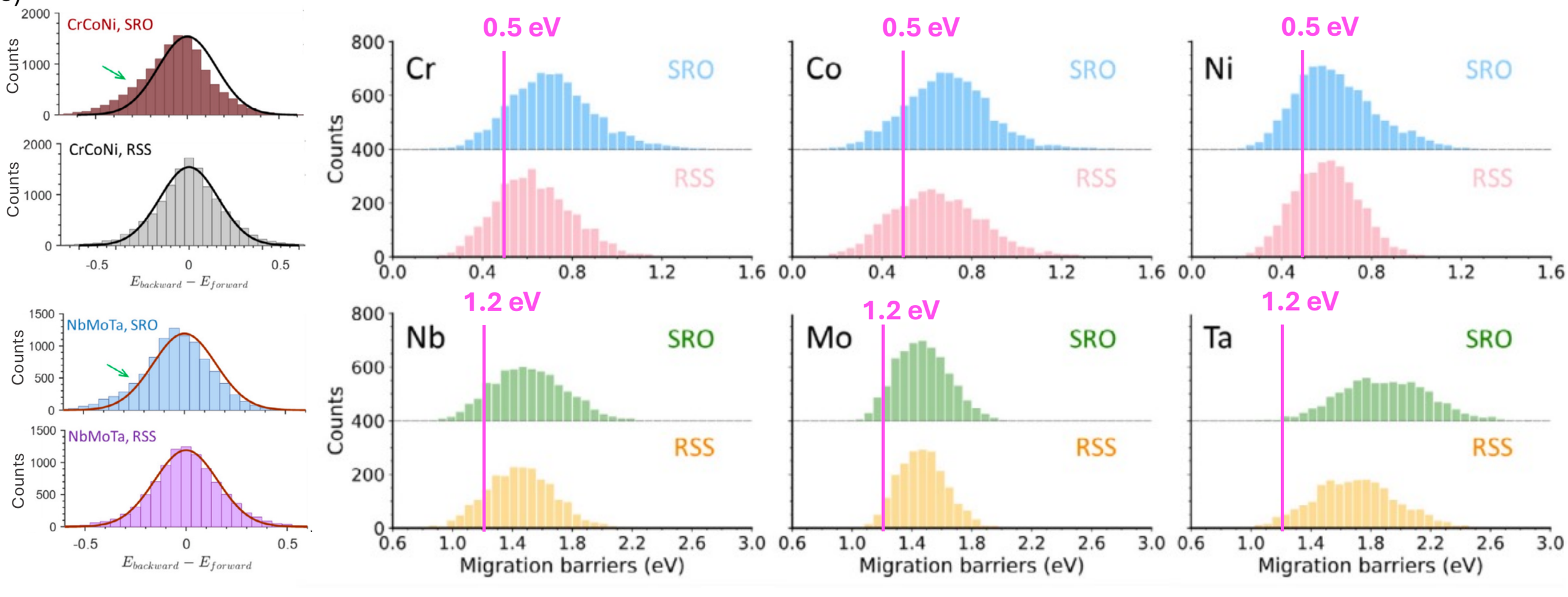
a)


Number of cascades
0
50
100
150
200
250
300
350
400
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
Defect Concentration
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
DPA
5
4
3
2
1
Al0.5CrCuFeNi
AlCrCuFeNi
AlCrCuFeNi2
AlCrCuFeNi3
CrCuFeNi

b)


0.6 eV
Count
200
100
0
5
AlCrCuFeNi
E_{ave}^{tot} = 0.974 ± 0.003
V ↔ Al
V ↔ Cr
V ↔ Cu
V ↔ Fe
V ↔ Ni
3
AlCrCuFeNi2
E_{ave}^{tot} = 1.0 ± 0.003
4
AlCrCuFeNi3
E_{ave}^{tot} = 1.01 ± 0.003
2
Al0.5CrCuFeNi
E_{ave}^{tot} = 0.991 ± 0.003
1
CrCuFeNi
E_{ave}^{tot} = 1.03 ± 0.003
0.0
0.5
1.0
1.5
2.0
2.5
Vacancy migration energy (eV)

c)


CrCoNi, SRO
CrCoNi, RSS
NbMoTa, SRO
NbMoTa, RSS
Counts
$E_{backward} - E_{forward}$
0.5 eV
Cr
Co
Ni
1.2 eV
Nb
Mo
Ta
SRO
RSS
Migration barriers (eV)